Credit Risk Default Prediction - Machine Learning Project Report

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Executive Summary

This project develops an end-to-end machine learning framework for predicting whether a loan applicant is likely to default within the next 12 months. Multiple datasets containing applicant information, credit bureau records, repayment history, previous loan behavior, and credit card usage were integrated to build a comprehensive credit risk prediction system.

The workflow included exploratory data analysis, feature engineering, data leakage prevention, model development, cross-validation, hyperparameter optimization, and model interpretation. Six machine learning models were evaluated and compared using multiple performance metrics.

Among all evaluated models, CatBoost achieved the strongest overall performance with a ROC-AUC of 0.8661 and KS Statistic of 0.5986, while maintaining strong interpretability through SHAP analysis.

1. Problem Statement

The objective of this project is to develop a machine learning model capable of predicting whether a loan applicant will default within the next twelve months.

The solution should:

- Accurately identify high-risk applicants.
- Handle class imbalance effectively.
- Provide interpretable predictions.
- Identify the key drivers of default risk.
- Support informed credit risk assessment.

2. Dataset Overview

The analytical dataset was constructed by integrating multiple sources of borrower information, including:

- Application data
- Credit bureau records
- Previous loan history
- Repayment behavior
- Credit card activity

Missing values were handled through median imputation where appropriate. Categorical variables were transformed using one-hot encoding, and numerical variables were prepared for modeling through feature scaling and transformation.

3. Feature Engineering

A significant portion of the project focused on transforming raw transactional and credit data into predictive variables.

Feature Categories

Category	Description
Ratio Features	Debt-to-income and loan-related ratios
Behavioral Features	Delinquency and repayment behavior metrics
Credit Utilization Features	Utilization stress and credit usage indicators
Time-Based Features	Account age, history length, and DPD metrics
Credit Burden Features	Outstanding balances and repayment gaps
Credit Profile Features	Bureau score and account-level credit indicators

Key Engineered Features

- Delinquency Rate
- Previous Default Rate
- Utilization Ratio
- Average Days Past Due (DPD)
- Debt per Account
- Repayment Ratio
- Outstanding Ratio
- Credit Used Percentage
- Late Payment Rate

These engineered variables significantly improved predictive performance across multiple models.

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4. Data Leakage Prevention

Several safeguards were implemented to prevent information leakage and ensure realistic model evaluation.

Measures Implemented

- Feature engineering used only information available before prediction.
- Target information was excluded from feature creation.
- Train-test separation was maintained throughout the workflow.
- Scaling was fitted only on training data.
- Cross-validation was performed exclusively on training data.
- Hyperparameter tuning avoided exposure to holdout test data.

These controls helped ensure reliable estimates of real-world performance.

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5. Modeling Approach

Handling Class Imbalance

The target variable was highly imbalanced, with approximately 8.6% of applicants belonging to the default class.

Instead of oversampling techniques, model-level weighting strategies were used:

- Logistic Regression: `class_weight='balanced'`
- Decision Tree: `class_weight='balanced'`
- Random Forest: `class_weight='balanced'`
- XGBoost: `scale_pos_weight`

This approach improved minority-class detection while preserving the original data distribution.

Models Evaluated

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. XGBoost
5. LightGBM
6. CatBoost

Cross-validation was performed using Stratified K-Fold validation to ensure stable performance estimates.

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6. Feature Importance and Model Interpretation

Interpretability was a major component of the project. Multiple techniques were used to understand the factors influencing default predictions.

Logistic Regression

Coefficient analysis showed that:

- Lower bureau scores increase default risk.
- Higher delinquency counts increase risk.
- Previous defaults increase future default probability.
- Higher utilization ratios increase risk.
- More late payments increase risk.

Decision Tree

The most influential variable was:

- Bureau Score

Other important predictors included:

- Previous Default Rate
- Average DPD
- Utilization Ratio
- Delinquency Rate
- Late Payment Behavior

Random Forest

Top predictors included:

- Bureau Score
- Delinquency Rate
- Delinquent Accounts
- Average DPD
- Debt per Account
- Utilization Ratio
- DPD per Loan

SHAP Analysis

SHAP analysis was performed for:

- XGBoost
- LightGBM
- CatBoost

The analysis consistently identified historical credit behavior as the strongest determinant of future default risk.

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7. Top Features Driving Default Risk

Across Logistic Regression coefficients, tree-based feature importance, and SHAP analysis, the following variables consistently emerged as the strongest predictors:

Rank	Feature
1	Bureau Score
2	Delinquency Rate
3	Delinquent Accounts
4	Average DPD
5	Utilization Ratio
6	Previous Default Rate
7	Late Payments
8	Number of Defaults

These variables capture borrower creditworthiness and repayment discipline.

A consistent appearance across multiple models increases confidence that they are genuine drivers of default behavior.

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8. Model Evaluation Results

Performance Comparison

Model	ROC-AUC	PR-AUC	KS Statistic
Logistic Regression	0.8634	0.4316	0.5948
Decision Tree	0.8233	0.3052	0.5493
Random Forest	0.8548	0.3880	0.5653
XGBoost	0.8574	0.3898	0.5839
LightGBM	0.8616	0.3981	0.5983
CatBoost	**0.8661**	**0.4297**	**0.5986**

ROC Curve Analysis

Key findings:

- All models significantly outperformed random classification.
- CatBoost achieved the highest ROC-AUC.
- Logistic Regression performed remarkably well despite being the simplest model.
- LightGBM and XGBoost produced competitive results.
- Ensemble methods consistently outperformed the single Decision Tree model.

Precision-Recall Analysis

Given the imbalanced target distribution, PR-AUC was also evaluated.

Key findings:

- Logistic Regression achieved the highest PR-AUC.
- CatBoost achieved nearly identical PR-AUC while maintaining superior ROC-AUC and KS performance.
- Boosting models consistently outperformed the tree baseline.

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9. Risk Band Analysis

Predicted probabilities were segmented into risk groups to better understand default concentration.

Risk Band	Probability Range	Approximate Default Rate
Very Low Risk	0.00 - 0.05	~1%
Low Risk	0.05 - 0.12	~3%
Medium Risk	0.12 - 0.25	~8%
High Risk	0.25 - 0.50	~18%
Very High Risk	0.50 - 1.00	~29%

The analysis demonstrates clear separation between low-risk and high-risk customer groups.

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10. Final Model Selection

CatBoost was selected as the final model because it:

- Achieved the highest ROC-AUC (0.8661).
- Achieved the highest KS Statistic (0.5986).
- Produced strong cross-validation performance.
- Maintained competitive PR-AUC.
- Provided robust interpretability through SHAP analysis.

The model demonstrated the best balance between predictive performance and explainability.

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11. Key Insights

Credit Behavior Dominates

Historical credit behavior is substantially more predictive than demographic variables such as age or income.

Bureau Score is Critical

Bureau score consistently emerged as the strongest predictor of default risk across all interpretation methods.

Delinquency Matters

Delinquency-related variables repeatedly ranked among the most important predictors.

Utilization Signals Financial Stress

High credit utilization ratios were strongly associated with increased default probability.

Feature Engineering Added Value

Engineered features consistently appeared among the most influential predictors, validating the effectiveness of the feature engineering strategy.

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12. Conclusion

This project successfully developed a complete machine learning pipeline for credit risk default prediction. The solution combines extensive feature engineering, robust validation, multiple model architectures, and advanced interpretation techniques.

Among the six evaluated models, CatBoost emerged as the strongest overall performer. The consistency of findings across Logistic Regression coefficients, tree-based feature importance, and SHAP analysis demonstrates that historical credit behavior and repayment discipline are the primary drivers of default risk.

The resulting framework provides an interpretable and scalable approach for identifying high-risk applicants and supporting data-driven credit risk assessment.